

The Relationship Between Unemployment and U.S. Real GDP Growth

Final Project: Hypothesis Testing for Machine Learning

Indicator: IBM Machine Learning Certificate

Course 1. Exploratory Data Analysis for Machine Learning

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Statement on the Use of Artificial Intelligence

This project was completed through a collaborative learning process between the author and ChatGPT. The author identified the research topic, conducted the analysis, implemented the Python code, and evaluated the results. ChatGPT provided support in refining the research question, structuring the hypotheses, identifying suitable data sources, and organizing the analytical workflow. This collaboration was used as an educational tool to strengthen the author's skills in exploratory data analysis, data preparation, hypothesis testing, and machine learning.

Revisions

Versions	Descriptions	Date
V1.0	Initial report	07/24/2026
V1.1	Corrected Welch's t-test results: $t = -0.258$ and $p = 0.796$	07/28/2026
V1.2	Revised the alternative hypothesis to match the two-sided statistical test	07/28/2026
V1.3	Added model-performance interpretation, including the negative test R^2	07/28/2026
V1.4	Added the analysis period and justification for the 5% unemployment threshold	07/28/2026
V1.5	Added links to the data sources in the References section	07/28/2026
V1.6	Revised the description in the section of Hypothesis Testing	07/28/2026
V1.7	Added the analysis period	07/28/2026

Agenda

- ❖ Define the research question.
- ❖ Download unemployment and GDP data
- ❖ Data Cleaning & Feature Engineering
- ❖ Convert monthly unemployment into quarterly averages.
- ❖ Merge it with quarterly GDP.
- ❖ Calculate GDP growth.
- ❖ Create the 5% unemployment indicator.
- ❖ Perform EDA.
- ❖ Hypothesis testing : Conduct a Welch independent-samples t-test.
- ❖ Build a simple linear regression model.
- ❖ Evaluate the model.
- ❖ Conclusion.

❖ Define the research question

What is the relationship between unemployment rate and U.S. real GDP growth?

- Independent variable (X): U.S. unemployment rate
- Dependent variable (y): U.S. real GDP Growth

Hypothesis Testing

Designed two groups:

1. Unemployment high ($\mathcal{U}_{\text{high}}$) = mean GDP growth when unemployment $\geq 5\%$
2. Unemployment low ($\mathcal{U}_{\text{low}}$) = mean GDP growth when unemployment $< 5\%$

Note: The **5% unemployment rate** was chosen as an assumed threshold to compare GDP growth under different unemployment conditions. This value is used solely for practicing hypothesis testing and can be modified depending on the research question, dataset, or economic circumstances.

The Hypothesis test:

$$H_0: \mathcal{U}_{\text{high}} = \mathcal{U}_{\text{low}}$$

(Clarification: Average real GDP growth is not significantly different between high- and low-unemployment period.)

$$H_a: \mathcal{U}_{\text{high}} \neq \mathcal{U}_{\text{low}}$$

(Clarification: Average real GDP growth differs between periods of high and low unemployment.)

❖ Download unemployment and GDP data

Unemployment

```
[2]: # unemployment

unemployment_url= "https://fred.stlouisfed.org/graph/fredgraph.csv?id=UNRATE"
df_unem = pd.read_csv(unemployment_url)
df_unem.head()
```

```
[2]:
```

	observation_date	UNRATE
0	1948-01-01	3.4
1	1948-02-01	3.8
2	1948-03-01	4.0
3	1948-04-01	3.9
4	1948-05-01	3.5

GDP

```
[3]: ## GDP

gdp_url= "https://fred.stlouisfed.org/graph/fredgraph.csv?id=GDPC1"
df_gdp= pd.read_csv(gdp_url)
df_gdp.head()
```

```
[3]:
```

	observation_date	GDPC1
0	1947-01-01	2182.681
1	1947-04-01	2176.892
2	1947-07-01	2172.432
3	1947-10-01	2206.452
4	1948-01-01	2239.682

❖ Data Cleaning & Feature Engineering

Understanding the Dataset

➤ Identify the Features in the data

Variable	Meaning
GDP	Total economic output
GDP Growth	Percentage change of GDP
Unemployment	Percentage of labor force unemployed

➤ Identify the Frequency of the Features

Variable	Frequency
GDP	Quarterly
GDP Growth	Quarterly
Unemployment	Monthly

➤ Identify the units of Features

Variable	Units
GDP	Billions of dollars
GDP Growth	Percentage change of GDP
Unemployment	Percent

❖ Data Cleaning & Feature Engineering

Evaluate features

➤ Statistical test

```
[15]: df.describe()
```

```
[15]:
```

	observation_date	GDPC1	GDP_Growth	UNRATE
count	264	264.000000	264.000000	264.000000
mean	1993-02-14 12:10:54.545454	11883.396110	0.738457	5.861689
min	1960-04-01 00:00:00	3470.278000	-7.876784	3.400000
25%	1976-09-08 00:00:00	6439.606500	0.346776	4.633333
50%	1993-02-15 00:00:00	10607.061000	0.758476	5.566667
75%	2009-07-24 00:00:00	16845.826000	1.128664	6.908333
max	2026-01-01 00:00:00	24180.419000	7.762279	13.000000
std	NaN	5952.695668	1.052468	1.690784

❖ Data Cleaning & Feature Engineering

Evaluate features

➤ Correlation test

```
[17]: df.corr()
```

```
[17]:
```

	observation_date	GDPC1	GDP_Growth	UNRATE
observation_date	1.000000	0.988725	-0.136171	-0.069846
GDPC1	0.988725	1.000000	-0.119337	-0.155775
GDP_Growth	-0.136171	-0.119337	1.000000	-0.121698
UNRATE	-0.069846	-0.155775	-0.121698	1.000000

❖ Data Cleaning & Feature Engineering

➤ .isnull().sum()

```
[4]: df_unem.isnull().sum()
```

```
[4]: observation_date    0
      UNRATE            1
      dtype: int64
```

```
[5]: df_unem["UNRATE"] = df_unem.UNRATE.fillna(df_unem.UNRATE.mean())
```

```
[6]: # Check the null values
```

```
df_unem.isnull().sum()
```

```
[6]: observation_date    0
      UNRATE            0
      dtype: int64
```

```
[7]: df_gdp.isnull().sum()
```

```
[7]: observation_date    0
      GDPC1              0
      dtype: int64
```

Covert observation_date to datetime

```
[8]: df_unem["observation_date"] = pd.to_datetime(df_unem["observation_date"])
      df_gdp["observation_date"] = pd.to_datetime(df_gdp["observation_date"])
```

Convert values to numeric

```
[9]: df_unem["UNRATE"] = pd.to_numeric(df_unem["UNRATE"], errors="coerce")
      df_gdp["GDPC1"] = pd.to_numeric(df_gdp["GDPC1"], errors="coerce")
```

Checking the missing data

```
[10]: print("Missing unemployment values:")
       print(df_unem.isna().sum())
```

```
print("\nMissing GDP values:")
print(df_gdp.isna().sum())
```

```
Missing unemployment values:
observation_date    0
UNRATE            0
dtype: int64
```

```
Missing GDP values:
observation_date    0
GDPC1              0
dtype: int64
```

❖ Convert monthly unemployment into quarterly averages

```
[12]: df_unem= df_unem.set_index ("observation_date")

df_unem_quarterly= (df_unem["UNRATE"].resample("QS").mean().reset_index())
# QS means "quarter-start frequency".
# Each quarterly value represents the average unemployment rate across the three months in that quarter.

df_unem_quarterly.head()
```

```
[12]:
```

	observation_date	UNRATE
0	1960-01-01	5.133333
1	1960-04-01	5.233333
2	1960-07-01	5.533333
3	1960-10-01	6.266667
4	1961-01-01	6.800000

❖ Merge it with quarterly GDP.

```
[15]: df = pd.merge(df_gdp, df_unem_quarterly, on="observation_date", how="inner")  
  
df = df.dropna().reset_index(drop=True)  
df.head()
```

```
[15]:
```

	observation_date	GDPC1	GDP_Growth	UNRATE
0	1960-04-01	3498.246	-0.538357	5.233333
1	1960-07-01	3515.385	0.489931	5.533333
2	1960-10-01	3470.278	-1.283131	6.266667
3	1961-01-01	3493.703	0.675018	6.800000
4	1961-04-01	3553.021	1.697855	7.000000

➤ **Analysis period: 1960 Q2 to 2026 Q1, based on the merged dataset.**

```
[18]: print(df["observation_date"].max())  
print(df["observation_date"].min())  
  
2026-01-01 00:00:00  
1960-04-01 00:00:00
```

❖ Calculate GDP growth

```
[17]: df.describe()
```

```
[17]:
```

	observation_date	GDPC1	GDP_Growth	UNRATE
count	264	264.000000	264.000000	264.000000
mean	1993-02-14 12:10:54.545454	11883.396110	0.738457	5.861689
min	1960-04-01 00:00:00	3470.278000	-7.876784	3.400000
25%	1976-09-08 00:00:00	6439.606500	0.346776	4.633333
50%	1993-02-15 00:00:00	10607.061000	0.758476	5.566667
75%	2009-07-24 00:00:00	16845.826000	1.128664	6.908333
max	2026-01-01 00:00:00	24180.419000	7.762279	13.000000
std	NaN	5952.695668	1.052468	1.690784

❖ Create the 5% unemployment indicator

```
[22]: df["High_unemployment"] = np.where(df["UNRATE"] >= 5, "At least 5%", "Below 5%")
df["Unemployment_Group"] = np.where(df["UNRATE"] >= 5, "At least 5%", "Below 5%")
df.head()
```

```
[22]:
```

	observation_date	GDPC1	GDP_Growth	UNRATE	High_unemployment	Unemployment_Group
0	1960-04-01	3498.246	-0.538357	5.233333	At least 5%	At least 5%
1	1960-07-01	3515.385	0.489931	5.533333	At least 5%	At least 5%
2	1960-10-01	3470.278	-1.283131	6.266667	At least 5%	At least 5%
3	1961-01-01	3493.703	0.675018	6.800000	At least 5%	At least 5%
4	1961-04-01	3553.021	1.697855	7.000000	At least 5%	At least 5%

❖ Perform EDA

1. Summary statistics by group

```
• [23]: group_summary= (  
        df.groupby("Unemployment_Group") ["GDP_Growth"].agg( ["count", "mean", "median", "std", "min", "max"] )  
    )  
  
group_summary
```

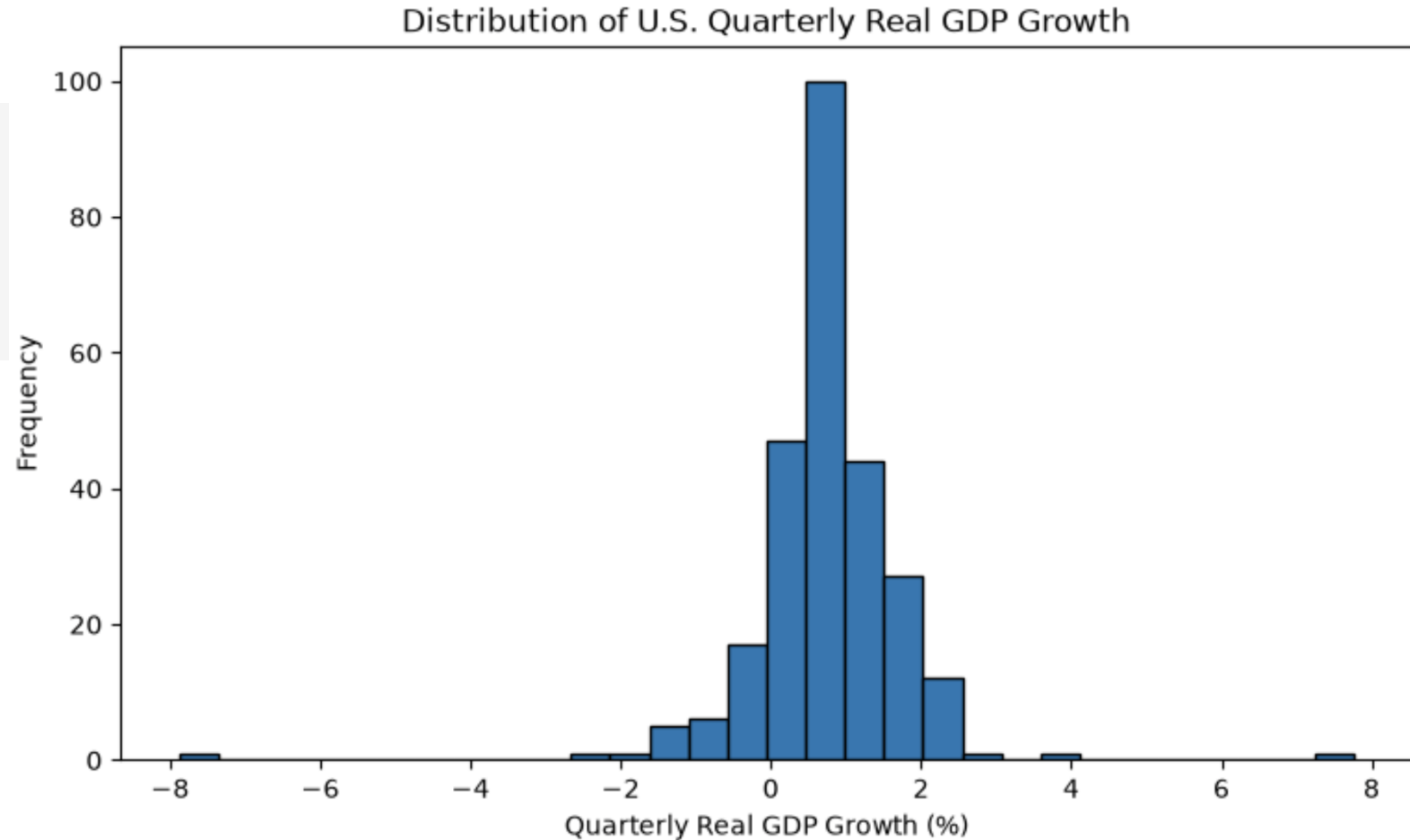
```
[23]:
```

		count	mean	median	std	min	max
Unemployment_Group							
	At least 5%	175	0.728393	0.774952	1.201429	-7.876784	7.762279
	Below 5%	89	0.758246	0.721900	0.675150	-1.316322	2.474942

❖ Perform EDA

2. Distribution of GDP Growth

```
plt.figure(figsize= (9,5))  
plt.hist(df["GDP_Growth"], bins = 30, edgecolor='black')  
plt.xlabel("Quarterly Real GDP Growth (%)")  
plt.ylabel("Frequency")  
plt.title("Distribution of U.S. Quarterly Real GDP Growth")  
plt.show()
```



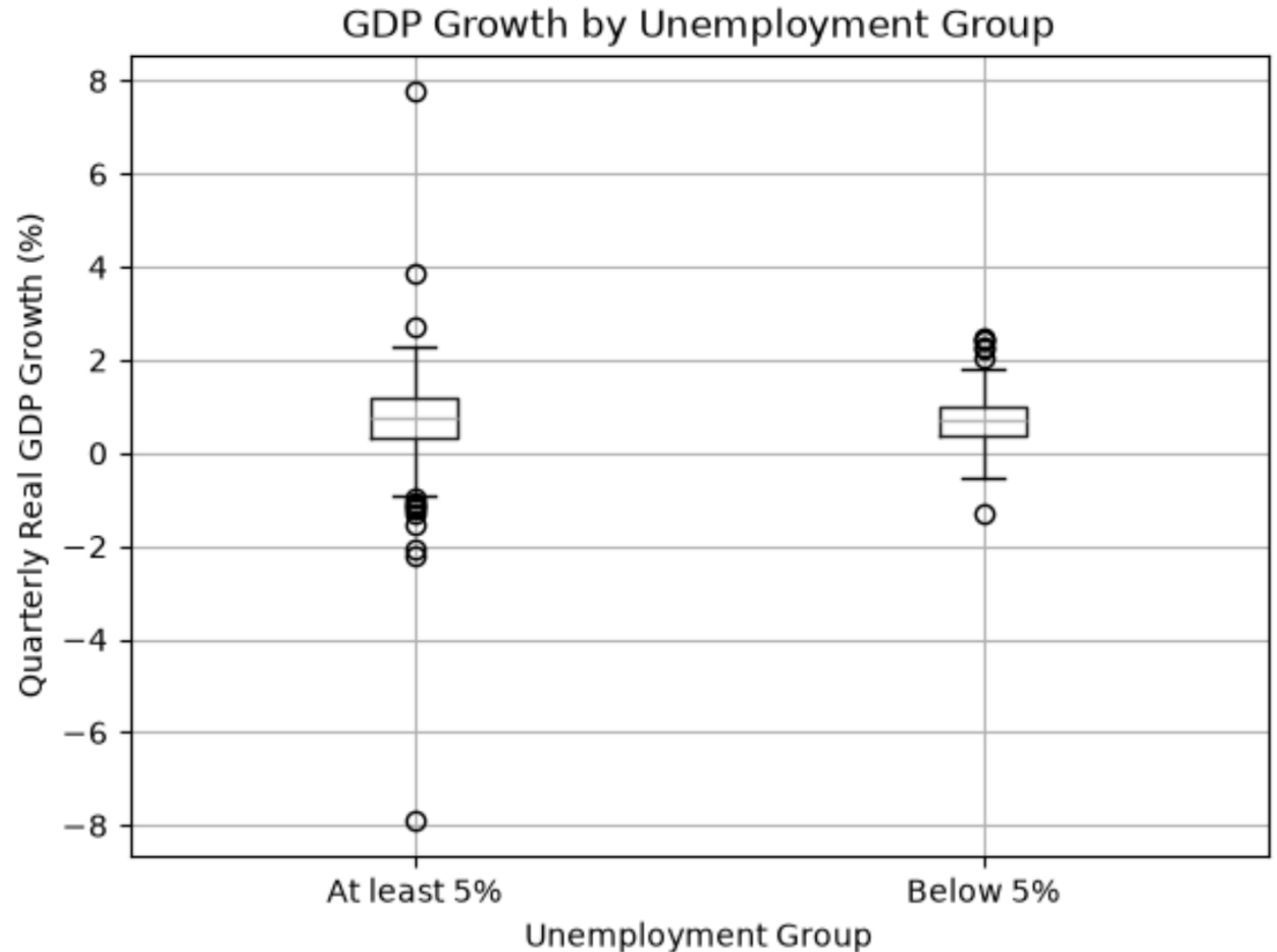
❖ Perform EDA

3. Boxplot by unemployment group

```
plt.figure(figsize=(8,5))

df.boxplot(column= "GDP_Growth",
            by= "Unemployment_Group")

plt.xlabel("Unemployment Group")
plt.ylabel("Quarterly Real GDP Growth (%)")
plt.title("GDP Growth by Unemployment Group")
plt.suptitle("")
plt.show()
```



❖ Perform EDA

4. Scatter plot

```
[74]: plt.figure(figsize=(9,5))

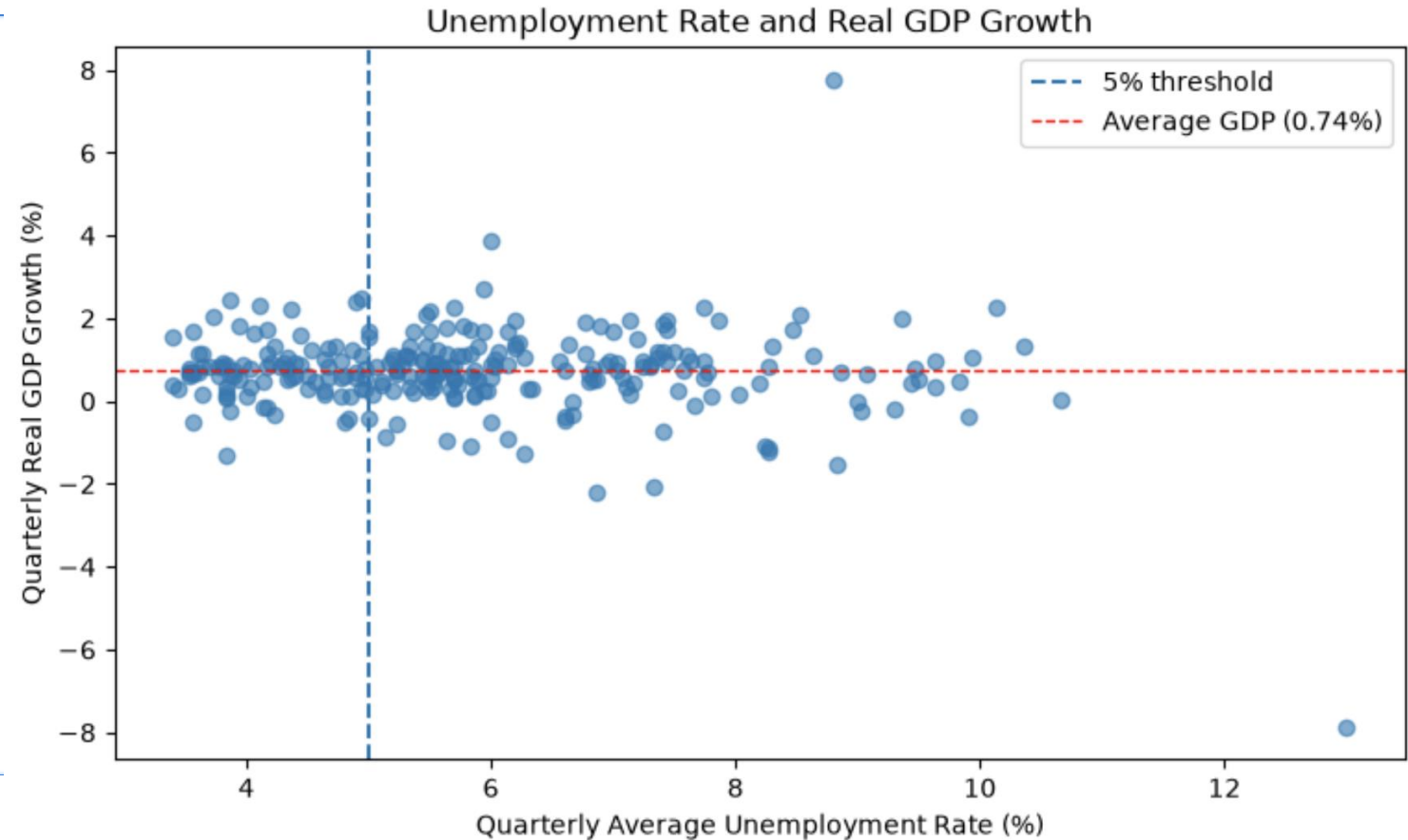
plt.scatter(df["UNRATE"],df["GDP_Growth"],alpha=0.6)

plt.axvline(x=5, linestyle= '--', label= "5% threshold")

avg_growth= df["GDP_Growth"].mean()
plt.axhline(
    y=avg_growth,
    color= 'r',
    linestyle= '--',
    linewidth= 1,
    label=f"Average GDP ({avg_growth:.2f}%)")
)

plt.xlabel("Quarterly Average Unemployment Rate (%)")
plt.ylabel("Quarterly Real GDP Growth (%)")
plt.title("Unemployment Rate and Real GDP Growth")
plt.legend()
plt.show()

print("Average GDP is ", avg_growth)
```



❖ Perform EDA

5. Time-series Plot

```
•[45]: fig, ax1 = plt.subplots(figsize= (12,6))

ax1.plot(
    df["observation_date"],
    df["UNRATE"],
    label= "Unemployment Rate",
    color='g'
)

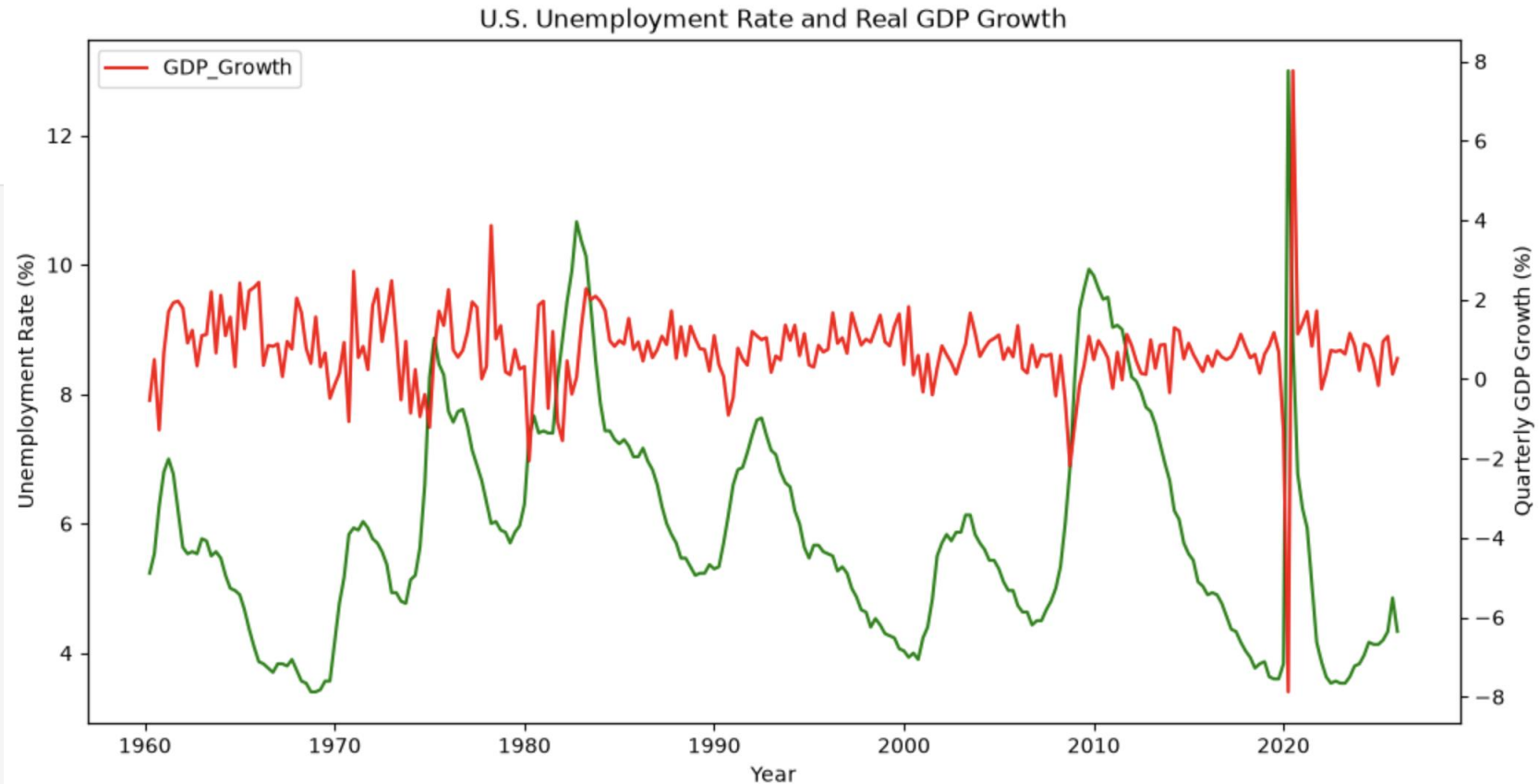
ax1.set_xlabel("Year")
ax1.set_ylabel("Unemployment Rate (%)")

ax2=ax1.twinx()

ax2.plot(df["observation_date"],
        df["GDP_Growth"],
        label= "GDP_Growth",
        color= 'red')

ax2.set_ylabel("Quarterly GDP Growth (%)")

plt.title("U.S. Unemployment Rate and Real GDP Growth")
plt.legend()
plt.show()
```



❖ Hypothesis testing : Conduct a Welch independent-samples t-test

```
[48]: high_unemployment = df.loc[df["UNRATE"]>= 5, "GDP_Growth"]
      low_unemployment = df.loc[df["UNRATE"]< 5, "GDP_Growth"]

print("High-unemployment observations:", len(high_unemployment))
print("Low-unemployment observations:", len(low_unemployment))
print("High-unemployment mean GDP growth:", high_unemployment.mean())
print("Low-unemployment mean GDP growth:", low_unemployment.mean())
```

```
High-unemployment observations: 175
Low-unemployment observations: 89
High-unemployment mean GDP growth: 0.7283930621216645
Low-unemployment mean GDP growth: 0.7582463498302441
```

Welch's t-test

```
[50]: t_stats, two_sided_p_value = stats.ttest_ind(
      high_unemployment,
      low_unemployment,
      equal_var=False,
      nan_policy = 'omit'
      )

print("T-statistic:", t_stats)
print("Two-sided p-value:", two_sided_p_value)

T-statistic: -0.2581838793341449
Two-sided p-value: 0.7964697154090858
```

Interpretation:

1. The negative t-statistic indicates that the mean GDP growth during periods of high unemployment is slightly lower than during periods of low unemployment.
2. The small absolute value of the t-statistic ($|t| = 0.258$) indicates that the observed difference in mean GDP growth between the two groups is very small relative to the sampling variability.
3. This t-test did not find statistically significant evidence of a difference in the means.
4. There is insufficient evidence that the two population means differ through the observation of p-value ($p=0.796>0.05$)

❖ Build a simple linear regression model

$$\text{GDP Growth} = \beta_0 + \beta_1 \text{Unemployment} + \epsilon$$

```
[63]: from sklearn.model_selection import train_test_split
      from sklearn.linear_model import LinearRegression
      from sklearn.metrics import (
          mean_absolute_error,
          mean_squared_error,
          r2_score
      )
```

```
[68]: X = df[["UNRATE"]]
      y = df["GDP_Growth"]

      X_train, X_test, y_train, y_test = train_test_split(
          X,
          y,
          test_size=0.20,
          random_state=42
      )

      model = LinearRegression()

      model.fit(X_train, y_train)

      y_pred = model.predict(X_test)
```

❖ Evaluate the model

```
[70]: mae= mean_absolute_error (y_test, y_pred)
      rmse= np.sqrt (mean_squared_error (y_test, y_pred))

      r2= r2_score (y_test, y_pred)

      print("Intercept: ", model.intercept_)
      print("Unemployment coefficient: ", model.coef_[0])
      print("MAE:", mae)
      print("RMSE:", rmse)
      print("R²", r2)

Intercept:  1.3959868033888019
Unemployment coefficient:  -0.11600145136145519
MAE: 0.708713260515871
RMSE: 1.2958508589436122
R² -0.07262889020653196
```

Interpretation:

Within this model, a one-percentage-point increase in unemployment is associated with approximately a 0.116-percentage-point decrease in quarterly real GDP growth.

The regression coefficient is negative, indicating that higher unemployment is associated with lower GDP growth. However, the model's test R^2 is -0.073, suggesting that unemployment alone has poor predictive power for quarterly GDP growth.

In fact, the model performs slightly worse than a constant prediction based on the average GDP growth in the test set.

This indicates that additional macroeconomic variables are needed to improve predictive performance.

Because this analysis uses time-series data, the random 80/20 train-test split was adopted for educational purposes. In practice, a chronological split would be more appropriate for real forecasting applications.

❖ Conclusion

The average quarterly real GDP growth rate was **0.758%** when unemployment was below 5% and **0.728%** when unemployment was at least 5%. Welch's t-test produced a **t-statistic of -0.258**, and the two-sided **p-value is approximately 0.796**. Therefore, at the 5% significance level, we **failed to reject the null hypothesis**.

The regression coefficient for unemployment was **-0.116**, indicating a **negative** association between unemployment and GDP growth. However, the analysis does not establish causality because other macroeconomic factors were not controlled.

References

- Source: Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis. Data accessed on July 24, 2026.

Source Link:

- GDPC1: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=GDPC1>
- UNRATE: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=UNRATE>